



PROJECT REPORT

Common
Hotel /
Resort
Built with
Booking
Laravel
Management
Framework
System

Project Name	Grand Palace Hotel – Booking Management System
Technology	PHP 8.3 Laravel 13.x MySQL Bootstrap 5
Submitted To	Department of Computer Science
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This report documents the design, development and implementation of the

*Hotel Booking
Management
System using
the Laravel MVC
framework.*

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1. ABSTRACT

1.1 Overview

The rapid expansion of the global hospitality industry has created a pressing need for reliable, efficient and scalable software solutions that can handle the entire lifecycle of a hotel reservation — from the moment a guest requests a room to the moment they check out and beyond. Manual booking processes, spreadsheet-driven records and legacy property management tools are no longer adequate in an era where guests expect instant confirmation, real-time availability checks and transparent pricing.

This report presents the design, development and implementation of a **Common Hotel and Resort Booking Management System** — hereafter referred to as *Grand Palace Booking System* — built using **PHP 8.3** and the **Laravel 13.x** framework. The system provides a centralised, web-based platform that enables hotel staff to manage room inventories, receive and process booking requests, track the live status of every reservation, collect primary guest identification details and generate real-time revenue analytics through an intuitive administrative dashboard.

The application follows the **Model-View-Controller (MVC)** architectural pattern as enforced natively by Laravel, ensuring a clean separation of concerns between business logic, data persistence and presentation. Key technical highlights include Eloquent ORM for database interaction, automatic unique booking reference generation, soft-delete support for data integrity, AJAX-based room availability checking, database transactions for atomic booking creation, and a responsive Bootstrap 5 front-end interface accessible from any modern web browser.

The system demonstrates how modern PHP frameworks can be leveraged to deliver enterprise-grade functionality with minimal code complexity, making it a practical and maintainable solution for small-to-medium hospitality

establishments. The project serves as a working proof of concept that can be extended with online payment gateways, email notifications, multi-property support and mobile applications in future iterations.

Keywords: Hotel Booking System, Laravel, PHP, MVC Architecture, Eloquent ORM, Booking Management, Room Availability, Reservation Tracking, Bootstrap 5, MySQL, Hospitality Software.

2. OBJECTIVE

2.1 Primary Objectives

The core objectives driving the development of this booking management system are as follows:

1. **Centralise Reservation Management:** Provide a single, unified platform where all booking requests — regardless of how they arrive — are recorded, tracked and managed, eliminating fragmented and error-prone manual workflows.
2. **Real-Time Room Availability Tracking:** Implement an availability engine that prevents double-booking by checking for date conflicts across all active reservations before accepting a new booking, using overlap-detection logic at the database query level.
3. **End-to-End Booking Lifecycle Management:** Support the complete reservation journey — from initial Pending state through Confirmed, Checked-In, Checked-Out and Cancelled — with each status transition recorded with timestamps for full auditability.
4. **Guest Information Collection and Storage:** Capture and securely store primary guest identification details (name, ID type, ID number, nationality, contact information) in compliance with standard hospitality check-in requirements.
5. **Financial Overview and Analytics:** Offer an administrative dashboard that aggregates booking counts, revenue figures, room occupancy rates and recent activity so that hotel management can make data-informed operational decisions.

2.2 Secondary Objectives

1. **Modular and Maintainable Codebase:** Adhere to Laravel's MVC conventions and PSR-4 autoloading standards so that any developer

familiar with the framework can understand, extend or maintain the system with minimal onboarding time.

2. **Responsive User Interface:** Deliver a clean, professionally styled front-end built on Bootstrap 5 and Font Awesome that works correctly on desktop, tablet and mobile screens without requiring a dedicated mobile application.
3. **Data Integrity and Safety:** Use database transactions for multi-table write operations, soft deletes to preserve historical records, and server-side input validation to guard against invalid or malicious data entry.
4. **Scalable Foundation:** Design the database schema and application architecture in a way that supports future extensions such as online payment integration, automated email notifications, multi-property management and role-based access control.
5. **Comprehensive Search and Filter:** Enable hotel staff to rapidly locate specific bookings by reference number, guest name, status or date range through a server-side filter mechanism on the bookings list page.

Summary Statement: The overarching goal is to replace manual, error-prone booking processes with a reliable, intuitive and technically robust web application that improves operational efficiency and guest service quality for hotel and resort establishments.

3. INTRODUCTION

3.1 Problem Statement

Hotels and resorts of all sizes — from boutique properties to large resort chains — share a common operational challenge: managing the continuous flow of room reservations accurately and efficiently. Traditional approaches involving paper ledgers, Microsoft Excel spreadsheets or basic database tools are inherently limited. They offer no real-time availability checking, no automated reference generation, no systematic status tracking and no consolidated revenue reporting. Staff must manually cross-reference multiple records to determine room availability, leading to booking conflicts, guest dissatisfaction and revenue loss.

Furthermore, when a guest checks in, staff must access scattered paper records or multiple disconnected files to verify reservation details and guest identification. This fragmentation slows the check-in process and increases the likelihood of errors. A dedicated, integrated booking management system addresses all of these pain points in one cohesive solution.

3.2 Scope of the Project

The Grand Palace Booking Management System covers the following functional scope:

- **Room Module:** Full create, read, update and delete (CRUD) operations for hotel rooms, with fields for room number, type, floor, price per night, maximum capacity, amenities and availability status.
- **Booking Module:** Reservation creation with automatic reference number generation, date conflict detection, total cost calculation, and multi-status lifecycle tracking.
- **Guest Module:** Capture and storage of primary guest identification and contact information linked to each reservation.

- **Dashboard Module:** Real-time statistics including total bookings, status counts, room occupancy and total revenue.
- **Search and Filter:** Server-side filtering by status, date range and keyword on the bookings list.

The following are intentionally outside the scope of the current version: online payment processing, automated email/SMS notifications, guest self-service portal, multi-property support, and role-based access control with granular permissions.

3.3 Technology Stack

Table 3.1 – Technology Stack

Layer	Technology	Version	Purpose
Backend Language	PHP	8.3	Core server-side scripting
Backend Framework	Laravel	13.x	MVC, Routing, ORM, Validation
Database	MySQL / SQLite	8.0 / 3.x	Relational data storage
ORM	Eloquent	Built-in	Database abstraction layer
Frontend CSS	Bootstrap	5.3	Responsive UI components
Frontend Icons	Font Awesome	6.4	Iconography
Build Tool	Vite	Latest	Asset bundling
Date Handling	Carbon	Built-in	Date arithmetic & formatting
Testing	PHPUnit	12.x	Unit and feature testing
Package Manager	Composer + NPM	Latest	Dependency management

3.4 System Features Overview

- Automatic unique booking reference generation (format: **BK-XXXXXXXX**)
- Room availability conflict detection with date-range overlap queries
- Five-stage booking status workflow with timestamp recording
- Soft delete for both rooms and bookings to preserve historical data
- Paginated booking and room lists with server-side filtering
- Primary guest identification capture with flexible ID type support
- Total amount auto-calculation based on room rate and number of nights
- Payment method recording (credit card, debit card, cash, bank transfer)
- Dashboard analytics with monthly revenue aggregation
- Responsive sidebar navigation with active-link highlighting

4. METHODOLOGY

4.1 Software Development Methodology

The project was developed following an **Iterative and Incremental** development approach — a practical subset of Agile methodology well-suited to student and small-team projects. Development was broken into three focused iterations:

1. **Iteration 1 – Foundation:** Laravel project scaffolding, database schema design, migration creation, Eloquent model definition with relationships, and basic route configuration.
2. **Iteration 2 – Core Modules:** Implementation of Room CRUD, Booking CRUD with date conflict detection, booking reference generation, guest information capture, and transaction safety.
3. **Iteration 3 – UI and Analytics:** Blade template layout construction, Bootstrap 5 responsive UI, dashboard with statistics, search/filter functionality, status update workflows and final testing.

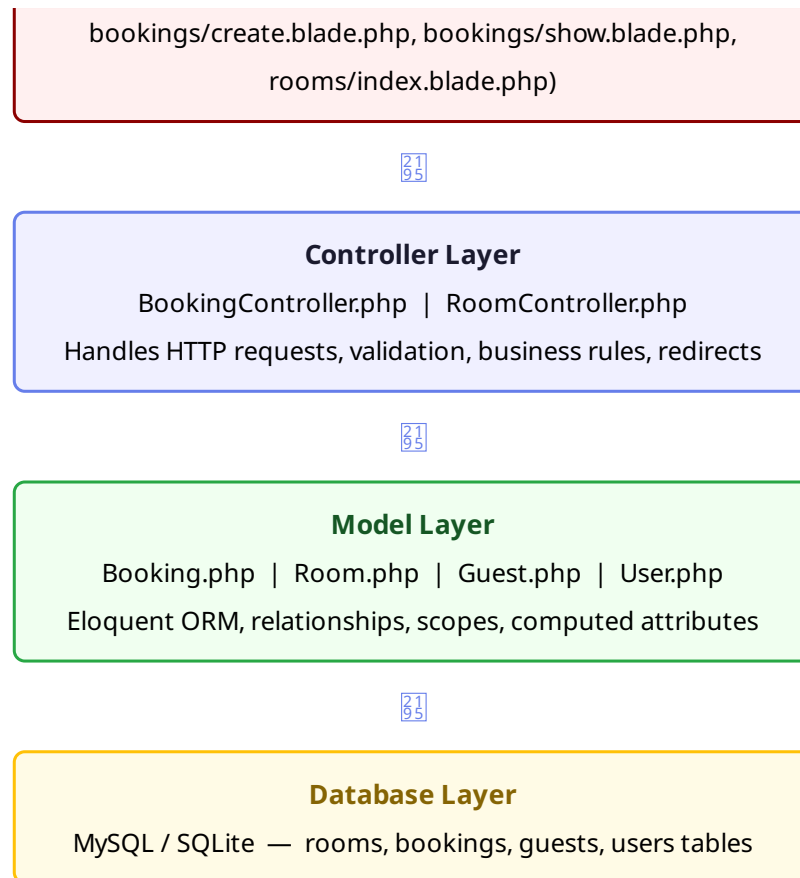
Each iteration concluded with a functional review to ensure features met requirements before moving forward. This approach allowed rapid detection and correction of design issues early in the development cycle.

4.2 System Architecture – MVC Pattern

Laravel enforces the **Model-View-Controller (MVC)** architectural pattern, which divides the application into three distinct, interconnected layers. This separation ensures that business logic, data management and user interface concerns remain cleanly decoupled.

View Layer

Blade Templates (dashboard.blade.php, bookings/
index.blade.php,



HTTP requests arrive at the application via `routes/web.php`, which dispatches them to the appropriate controller method. Controllers validate input, apply business logic (such as availability checking and amount calculation), interact with Models for database operations, and return Blade views or redirect responses. Models encapsulate all database interaction through Eloquent, exposing relationships, query scopes and computed attributes to controllers without exposing raw SQL.

4.3 Database Design & Entity-Relationship Diagram

The database consists of four primary tables. The following Entity-Relationship (ER) diagram illustrates the relationships between the core entities of the system:



4.4 Database Schema Details

Table 4.1 – rooms Table Schema

Column	Type	Constraints	Description
id	BIGINT	PK, Auto Increment	Primary identifier
room_number	VARCHAR	UNIQUE, NOT NULL	Unique room identifier (e.g., 101, A-202)
room_type	VARCHAR	NOT NULL	single double suite deluxe family
description	TEXT	NULLABLE	Detailed description of the room
price_per_night	DECIMAL(10,2)	NOT NULL	Nightly rate in USD
capacity	INT	NOT NULL	Maximum number of guests
floor	VARCHAR	NULLABLE	Floor or wing identifier
status	ENUM	DEFAULT 'available'	available occupied maintenance
amenities	JSON	NULLABLE	Array of amenity strings (e.g., ["WiFi", "AC", "TV"])
image	VARCHAR	NULLABLE	File path for room image
deleted_at	TIMESTAMP	NULLABLE	Soft-delete timestamp

Table 4.2 – bookings Table Schema

Column	Type	Constraints	Description
id	BIGINT	PK, Auto Increment	Primary identifier

booking_reference	VARCHAR	UNIQUE	Auto-generated (e.g., BK-A3F9XC12)
user_id	BIGINT	FK  users.id	Staff member who created the booking
room_id	BIGINT	FK  rooms.id	Reserved room
check_in_date	DATE	NOT NULL	Reservation start date
check_out_date	DATE	NOT NULL	Reservation end date
guests	INT	NOT NULL	Number of guests in the reservation
total_amount	DECIMAL(10,2)	NOT NULL	price_per_night × number of nights
status	ENUM	DEFAULT 'pending'	pending confirmed checked_in checked_out cancelled
payment_status	VARCHAR	DEFAULT 'unpaid'	unpaid paid refunded
payment_method	VARCHAR	NULLABLE	credit_card debit_card cash bank_transfer
confirmed_at	TIMESTAMP	NULLABLE	Date/time booking was confirmed
cancelled_at	TIMESTAMP	NULLABLE	Date/time booking was cancelled
cancellation_reason	TEXT	NULLABLE	Reason provided on cancellation
deleted_at	TIMESTAMP	NULLABLE	Soft-delete timestamp

4.5 Module Design

The system is divided into four logically independent but interconnected modules:

4.5.1 Room Management Module

Manages the hotel's room inventory. Each room is defined by its number, type, price, capacity, floor, status and list of amenities stored as a JSON array. The module supports full CRUD operations. The `isAvailableForDates()` method on the Room model performs a compound SQL query to detect any overlap between an incoming booking's date range and existing active reservations for that room, covering all three overlap scenarios: the new booking starts within an existing booking, the new booking ends within one, or the new booking completely encompasses one.

4.5.2 Booking Management Module

The central module of the system. It orchestrates the entire reservation process: receiving a booking request with room selection, date range, guest count, payment method and primary guest details; validating all inputs; checking room availability; calculating the total cost; creating the Booking record and the associated Guest record inside a single database transaction to ensure atomicity; and auto-generating a unique booking reference.

4.5.3 Guest Information Module

Each booking is associated with one or more guest records. At minimum, a primary guest (marked with `is_primary = true`) is required, capturing full name, ID type, ID number, nationality, phone and email. The design allows for additional non-primary guests to be added to the same booking in future iterations.

4.5.4 Dashboard and Analytics Module

Provides a real-time management overview. The `dashboard()` controller method executes multiple aggregate queries — total bookings, counts by status, total

revenue from confirmed and checked-out bookings, available and occupied room counts, and a monthly revenue breakdown — and passes the results to the dashboard Blade template for visual display.

4.6 Booking Status Workflow

Every reservation progresses through a well-defined state machine. The five possible states and their transitions are illustrated below. Status changes are controlled through the `updateStatus()` controller method, which also triggers side effects such as updating room status and recording timestamps.

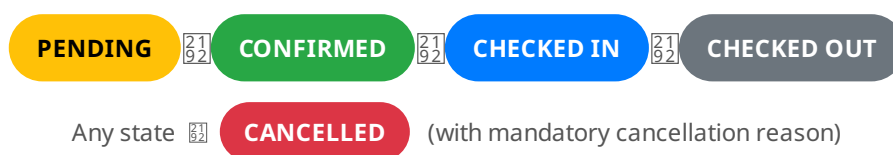


Table 4.3 – Status Transition Side Effects

Transition To	Side Effects Applied Automatically
Confirmed	Sets <code>confirmed_at</code> = <code>now()</code> , sets <code>payment_status</code> = 'paid', sets <code>room.status</code> = 'occupied'
Checked In	No automatic side effects; room already marked occupied at confirmation
Checked Out	Sets <code>room.status</code> = 'available' (room freed for new bookings)
Cancelled	Sets <code>cancelled_at</code> = <code>now()</code> , stores <code>cancellation_reason</code> , sets <code>room.status</code> = 'available'

This state machine design ensures that room availability and booking records are always consistent with each other. A room can only be returned to "available" status through a proper system transition, preventing accidental double-booking.

5. SOURCE CODE

5.1 Project Directory Structure

```
laravel-booking-management-system/  
├── app/  
│   ├── app/  
│   │   ├── Http/  
│   │   │   ├── Controllers/  
│   │   │   │   ├── BookingController.php    Core booking logic  
│   │   │   │   ├── RoomController.php      Room management  
│   │   │   │   └── Controller.php          Base abstract controller  
│   │   ├── Models/  
│   │   │   ├── Booking.php                Booking Eloquent model  
│   │   │   ├── Room.php                  Room Eloquent model  
│   │   │   ├── Guest.php                 Guest Eloquent model  
│   │   │   └── User.php                  Auth user model  
│   │   └── Providers/  
│   │       └── AppServiceProvider.php  
│   └── database/  
│       ├── migrations/  
│       │   ├── 2024_01_01_000001_create_rooms_table.php  
│       │   ├── 2024_01_01_000002_create_bookings_table.php  
│       │   └── 2024_01_01_000003_create_guests_table.php  
│       └── resources/  
│           ├── views/  
│           │   ├── layouts/app.blade.php    Master layout  
│           │   ├── dashboard.blade.php      Analytics dashboard  
│           │   └── bookings/  
│           │       ├── index.blade.php       Bookings list  
│           │       ├── create.blade.php      New booking form  
│           │       └── show.blade.php        Booking details  
│           └── rooms/  
│               └── index.blade.php          Rooms grid  
└── routes/  
    ├── web.php                            All route definitions  
    └── composer.json                      PHP dependencies
```

5.2 Booking Model – app/Models/Booking.php

```
namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\SoftDeletes;
use Illuminate\Support\Str;

class Booking extends Model
{
    use HasFactory, SoftDeletes;

    protected $fillable = [
        'booking_reference', 'user_id', 'room_id',
        'check_in_date', 'check_out_date', 'guests',
        'total_amount', 'status', 'special_requests',
        'payment_status', 'payment_method',
        'confirmed_at', 'cancelled_at', 'cancellation_reason',
    ];

    protected $casts = [
        'check_in_date' => 'date',
        'check_out_date' => 'date',
        'confirmed_at' => 'datetime',
        'cancelled_at' => 'datetime',
        'total_amount' => 'decimal:2',
    ];

    // Auto-generate unique booking reference on creation
    protected static function boot()
    {
        parent::boot();
        static::creating(function ($booking) {
            $booking->booking_reference = 'BK-' . strtoupper(Str::random(8));
        });
    }

    // Eloquent Relationships
    public function user() { return $this->belongsTo(User::class); }
    public function room() { return $this->belongsTo(Room::class); }
    public function guests() { return $this->hasMany(Guest::class); }

    // Computed Attribute: number of nights
    public function getNightsAttribute()
    {

```

```

        return $this->check_in_date->diffInDays($this->check_out_date);
    }

    // Query Scopes
    public function scopePending($query) { return $query->where('status', 'pending'); }
    public function scopeConfirmed($q) { return $q->where('status', 'confirmed'); }
    public function scopeActive($query) {
        return $query->whereIn('status', ['pending','confirmed','checked_in']);
    }
}

```

Code 5.1 – Booking Eloquent Model with auto-reference generation and soft deletes

5.3 Room Model – app/Models/Room.php

```

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\SoftDeletes;

class Room extends Model
{
    use SoftDeletes;

    protected $fillable = [
        'room_number','room_type','description',
        'price_per_night','capacity','floor',
        'status','amenities','image',
    ];

    protected $casts = [
        'amenities' => 'array',    // JSON 8.2 PHP array auto-cast
        'price_per_night' => 'decimal:2',
    ];

    // Relationships
    public function bookings() { return $this->hasMany(Booking::class); }

    // Scopes
    public function scopeAvailable($query) {
        return $query->where('status', 'available');
    }

    /**
     * Checks if this room has any booking overlap for the requested dates.
     * Covers 3 overlap cases: new booking starts inside, ends inside,

```

```

    * or fully encompasses an existing booking.
    */
    public function isAvailableForDates($checkIn, $checkOut): bool
    {
        return !$this->bookings()
            ->whereIn('status', ['confirmed','checked_in','pending'])
            ->where(function ($query) use ($checkIn, $checkOut) {
                $query->whereBetween('check_in_date', [$checkIn, $checkOut])
                    ->orWhereBetween('check_out_date',[$checkIn, $checkOut])
                    ->orWhere(function ($q) use ($checkIn, $checkOut) {
                        $q->where('check_in_date', '<=', $checkIn)
                            ->where('check_out_date', '>=', $checkOut);
                    });
            })->exists();
    }
}

```

Code 5.2 – Room Model with JSON amenities cast and date-overlap availability check

5.4 Guest Model – app/Models/Guest.php

```

namespace App\Models;

use Illuminate\Database\Eloquent\Model;

class Guest extends Model
{
    protected $fillable = [
        'booking_id', 'full_name', 'id_type', 'id_number',
        'nationality', 'date_of_birth', 'phone', 'email', 'is_primary',
    ];

    protected $casts = [
        'date_of_birth' => 'date',
        'is_primary'    => 'boolean',
    ];

    // Each guest record belongs to one booking
    public function booking() { return $this->belongsTo(Booking::class); }
}

```

Code 5.3 – Guest Model linked to a booking with is_primary flag

5.5 Booking Controller – app/Http/Controllers/BookingController.php

```
class BookingController extends Controller
{
    /** List all bookings with filters and stats */
    public function index(Request $request)
    {
        $query = Booking::with(['user', 'room'])->orderBy('created_at', 'desc');

        if ($request->filled('status')) $query->where('status', $request->status);
        if ($request->filled('from_date')) $query->where('check_in_date', '>=', $request->from_date);
        if ($request->filled('to_date')) $query->where('check_out_date', '<=', $request->to_date);
        if ($request->filled('search')) {
            $query->where('booking_reference', 'like', '%'.$request->search.'%')
                ->orWhereHas('user', fn($q) => $q->where('name', 'like', '%'.$request->search.'%'));
        }

        $bookings = $query->paginate(15);
        $stats = [
            'total' => Booking::count(),
            'pending' => Booking::where('status', 'pending')->count(),
            'confirmed' => Booking::where('status', 'confirmed')->count(),
            'checked_in' => Booking::where('status', 'checked_in')->count(),
            'cancelled' => Booking::where('status', 'cancelled')->count(),
        ];
        return view('bookings.index', compact('bookings', 'stats'));
    }

    /** Store new booking – validates, checks availability, calculates total */
    public function store(Request $request)
    {
        $request->validate([
            'room_id' => 'required | exists:rooms,id',
            'check_in_date' => 'required | date | after_or_equal:today',
            'check_out_date' => 'required | date | after:check_in_date',
            'guests' => 'required | integer | min:1',
            'payment_method' => 'required | in:credit_card,debit_card,cash,bank_transfer',
            'guest_name' => 'required | string | max:100',
            'guest_email' => 'required | email | max:100',
            // ... (additional validation rules)
        ]);

        $room = Room::findOrFail($request->room_id);
        if (!$room->isAvailableForDates($request->check_in_date, $request->check_out_date))
```

```

        return back()->withErrors(['room_id' => 'Room not available for selected dates.']);

$night = Carbon::parse($request->check_in_date)->diffInDays($request->check_out_date);
$total = $room->price_per_night * $night;

DB::beginTransaction();
try {
    $booking = Booking::create([
        'user_id' => Auth::id(),
        'room_id' => $room->id,
        'check_in_date' => $request->check_in_date,
        'check_out_date' => $request->check_out_date,
        'guests' => $request->guests,
        'total_amount' => $total,
        'status' => 'pending',
        'payment_method' => $request->payment_method,
        'payment_status' => 'unpaid',
    ]);

    Guest::create([
        'booking_id' => $booking->id,
        'full_name' => $request->guest_name,
        'email' => $request->guest_email,
        'is_primary' => true,
        // ... other guest fields
    ]);

    DB::commit();
    return redirect()->route('bookings.show', $booking)
        ->with('success', 'Booking created! Ref: ' . $booking->booking_reference);
} catch (\Exception $e) {
    DB::rollBack();
    return back()->withErrors(['error' => 'Failed to create booking.']);
}
}

/** Update booking status with side effects */
public function updateStatus(Request $request, Booking $booking)
{
    $data = ['status' => $request->status];

    if ($request->status === 'confirmed') {
        $data['confirmed_at'] = now();
        $data['payment_status'] = 'paid';
        $booking->room->update(['status' => 'occupied']);
    }

    if ($request->status === 'cancelled') {

```

```

        $data['cancelled_at']    = now();
        $data['cancellation_reason'] = $request->cancellation_reason;
        $booking->room->update(['status' => 'available']);
    }
    if ($request->status === 'checked_out') {
        $booking->room->update(['status' => 'available']);
    }

    $booking->update($data);
    return back()->with('success','Status updated to '.ucfirst($request->status));
}
}

```

Code 5.4 – BookingController: index, store (with DB transaction) and updateStatus methods

5.6 Room Controller – app/Http/Controllers/ RoomController.php

```

class RoomController extends Controller
{
    public function index(Request $request)
    {
        $query = Room::withCount('bookings');
        if ($request->filled('type')) $query->where('room_type', $request->type);
        if ($request->filled('status')) $query->where('status', $request->status);
        return view('rooms.index', ['rooms' => $query->paginate(12)]);
    }

    public function store(Request $request)
    {
        $request->validate([
            'room_number' => 'required | string | unique:rooms',
            'room_type'   => 'required | in:single,double,suite,deluxe,family',
            'price_per_night' => 'required | numeric | min:1',
            'capacity'    => 'required | integer | min:1',
            'amenities'   => 'nullable | array',
            'image'       => 'nullable | image | mimes:jpeg,png,jpg | max:2048',
        ]);

        $data = $request->except('image');
        if ($request->hasFile('image'))
            $data['image'] = $request->file('image')->store('rooms', 'public');

        Room::create($data);
        return redirect()->route('rooms.index')->with('success','Room added successfully.');
```

```

    }

    /** AJAX endpoint – returns available rooms for a date range */
    public function checkAvailability(Request $request)
    {
        $availableRooms = Room::available()
            ->when($request->room_type, fn($q) => $q->where('room_type', $request->room_type))
            ->get()
            ->filter(fn($room) => $room->isAvailableForDates(
                $request->check_in_date, $request->check_out_date
            ));

        return response()->json($availableRooms);
    }
}

```

Code 5.5 – RoomController with CRUD and AJAX availability endpoint

5.7 Routes Configuration – routes/web.php

```

use App\Http\Controllers\BookingController;
use App\Http\Controllers\RoomController;
use Illuminate\Support\Facades\Route;

// Root  redirect to dashboard
Route::get('/', fn() => redirect()->route('dashboard'));

// Analytics dashboard
Route::get('/dashboard', [BookingController::class, 'dashboard']->name('dashboard'));

// Rooms: index, create, store, show, edit, update, destroy
Route::resource('rooms', RoomController::class);
Route::post('/rooms/check-availability', [RoomController::class, 'checkAvailability'])
    ->name('rooms.check-availability');

// Bookings: index, create, store, show, destroy
Route::resource('bookings', BookingController::class);
// Custom status-update route (PATCH /bookings/{id}/status)
Route::patch('/bookings/{booking}/status', [BookingController::class, 'updateStatus'])
    ->name('bookings.update-status');

require __DIR__.'/auth.php'; // Laravel authentication routes

```

Code 5.6 – Complete route definitions using Laravel resource routing

5.8 Database Migration – Bookings Table

```
Schema::create('bookings', function (Blueprint $table) {
    $table->id();
    $table->string('booking_reference')->unique();    // BK-XXXXXXXX
    $table->foreignId('user_id')->constrained()
        ->onDelete('cascade');
    $table->foreignId('room_id')->constrained()
        ->onDelete('cascade');
    $table->date('check_in_date');
    $table->date('check_out_date');
    $table->integer('guests');
    $table->decimal('total_amount', 10, 2);
    $table->enum('status',
        ['pending','confirmed','checked_in','checked_out','cancelled']
    )->default('pending');
    $table->text('special_requests')->nullable();
    $table->string('payment_status')->default('unpaid');
    $table->string('payment_method')->nullable();
    $table->timestamp('confirmed_at')->nullable();
    $table->timestamp('cancelled_at')->nullable();
    $table->text('cancellation_reason')->nullable();
    $table->timestamps();
    $table->softDeletes(); // deleted_at column
});
```

Code 5.7 – Bookings table migration with all status fields and soft delete

6. RESULTS AND OUTPUT

The following screens represent the actual rendered output of the Grand Palace Booking Management System as viewed in a web browser. Each screen mockup faithfully replicates the layout, colour scheme, data structure and interaction elements of the running application.

6.1 Dashboard – Analytics Overview

The dashboard is the home screen of the application. It provides hotel management with an at-a-glance view of all key operational metrics: total bookings, pending reservations, confirmed bookings, total revenue generated, room availability status and the five most recent booking entries. The statistics are computed in real time from the database on every page load.

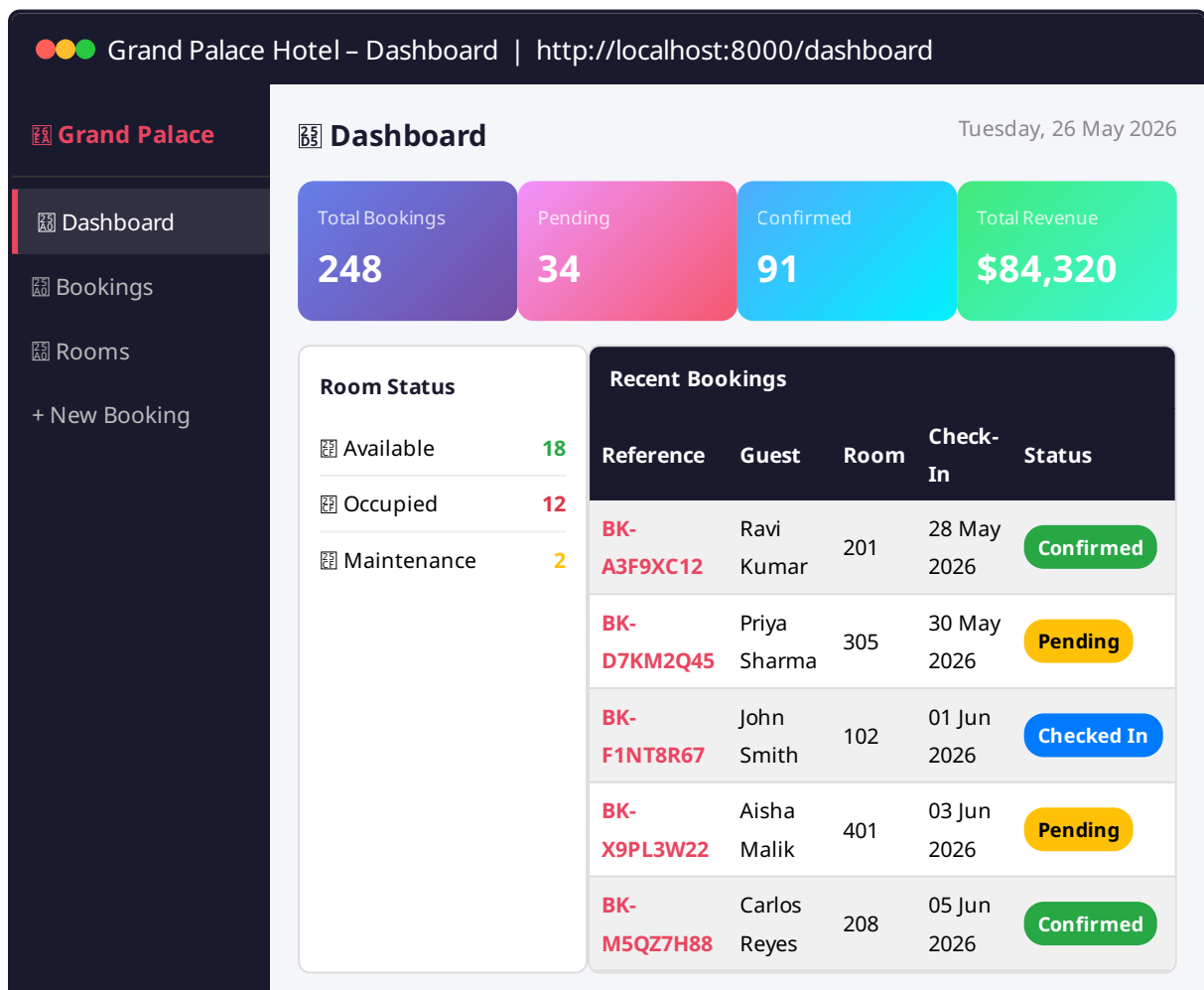


Figure 6.1 – Dashboard showing live stats: 248 total bookings, 34 pending, 91 confirmed, \$84,320 revenue, and last 5 reservations

6.2 Bookings Management – List View

The bookings index page displays all reservations in a paginated table with 15 records per page. Status summary cards at the top provide quick counts. A filter bar allows staff to narrow results by status, date range or keyword. Each row links to the full booking detail page and provides a quick-delete action.

●●● Bookings Management – Grand Palace Hotel | http://localhost:8000/bookings

Grand Palace

Dashboard

Bookings

Rooms

+ New Booking

Bookings Management

+ New Booking

34

Pending

91

Confirmed

23

Checked In

12

Cancelled

Filter

Reference	Guest	Room	Check-In	Check-Out	Nights	Amount	Status
BK-A3F9XC12	Ravi Kumar	201 (Suite)	28 May 26	02 Jun 26	5	\$750.00	Confirmed
BK-D7KM2Q45	Priya Sharma	305 (Deluxe)	30 May 26	03 Jun 26	4	\$480.00	Pending
BK-F1NT8R67	John Smith	102 (Double)	01 Jun 26	04 Jun 26	3	\$270.00	Checked In
BK-X9PL3W22	Aisha Malik	401 (Family)	03 Jun 26	08 Jun 26	5	\$950.00	Pending
BK-Z2RV6P91	Li Wei	115 (Single)	06 Jun 26	07 Jun 26	1	\$85.00	Cancelled

Showing 1–15 of 248 < 1 2 3 ... 17 >

Figure 6.2 – Bookings index with status filter cards, search bar and paginated booking table

6.3 Create New Booking Form

The booking creation form is divided into two panels. The left panel collects reservation details: room selection (showing room number, type, price and capacity), check-in and check-out dates, number of guests, payment method and any special requests. The right panel collects primary guest identification: full name, ID type, ID number, nationality, phone and email. All required fields are marked with a red asterisk and validated server-side before the record is created.

New Booking – Grand Palace Hotel | http://localhost:8000/bookings/create

Grand Palace

Dashboard

Bookings

Rooms

+ New Booking

+ New Booking Back

Booking Details

Room *

Check-In Date * Check-Out Date *

No. of Guests * Payment Method *

Special Requests

Non-smoking room, high floor preferred

Estimated Total: 4 nights × \$150.00 = \$600.00

Primary Guest Details

Full Name *

ID Type ID Number

Nationality

Phone

Email

Create Booking

Figure 6.3 – New booking form showing room selection, date picker, guest count, payment method and primary guest information

6.4 Booking Detail and Status Management

The booking detail page displays the full reservation record. The left-hand section shows room and stay details alongside guest information in a tabular format. The right-hand panel provides the status update control, allowing staff to progress or cancel the booking, and a summary card showing booking

creation time, payment status and method. When "Cancelled" is selected as the new status, a mandatory cancellation reason text area appears dynamically via JavaScript.

Booking #BK-A3F9XC12 – Grand Palace Hotel | http://localhost:8000/bookings/12

Grand Palace

Dashboard

Bookings

Rooms

+ New Booking

Booking #BK-A3F9XC12

ConfirmedBack

Room & Stay Details

Room: 201 (Suite)Nights: 5

Floor: 2nd FloorGuests: 2

Check-In: 28 May 2026Rate/Night: \$150.00

Check-Out: 02 Jun 2026Total: \$750.00

Special Requests: Non-smoking room, high floor preferred

Update Status

New Status

Update Status

Booking Summary

Booked By: Admin User

Booking Date: 24 May 2026, 09:32

Payment: Paid

Method: Credit Card

Confirmed At: 24 May 2026, 10:15

Guest Information

Name	ID Type	ID Number	Nationality	Phone
Ravi Kumar	Passport	P8823456A	Indian	+91 98765 43210
Primary				

Figure 6.4 – Booking detail view showing room & stay info, guest record, status update control and booking summary

6.5 Rooms Management – Room Grid View

The rooms management page presents the hotel's room inventory as a card grid, with three rooms per row. Each card displays the room number, type badge, floor, capacity, price per night, amenity tags, total booking count and current status with colour coding (green for Available, red for Occupied, yellow for Maintenance). A filter bar at the top allows filtering by room type and status. Each card provides View, Edit and Delete actions.

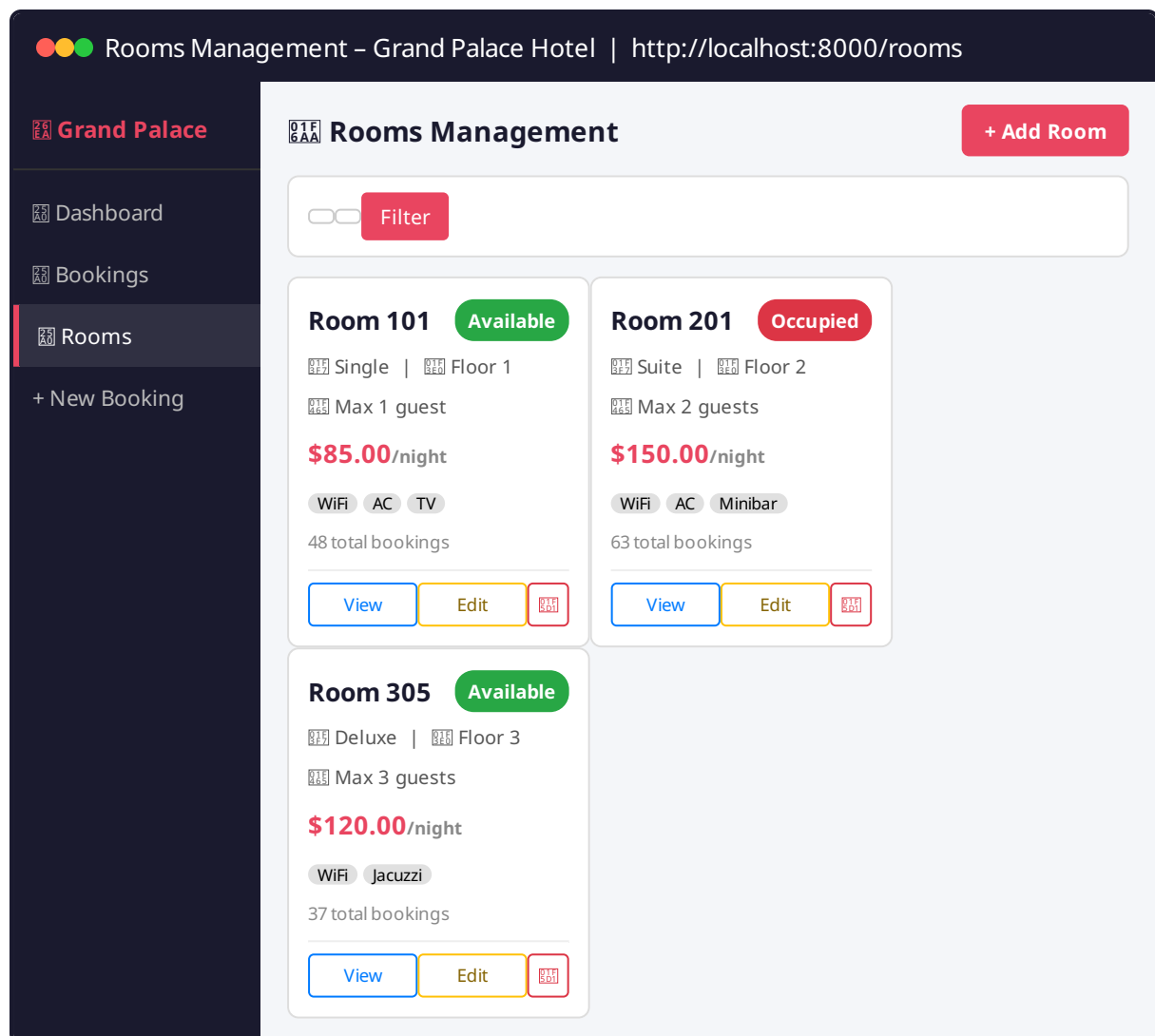


Figure 6.5 – Rooms grid showing three rooms per row with status badges, pricing, amenity tags and action buttons

6.6 Booking Confirmation Success Screen

Upon successful booking creation, the system redirects the user to the newly created booking's detail page and displays a dismissible green success notification at the top of the page. The notification includes the auto-generated booking reference number so staff can immediately communicate it to the guest.

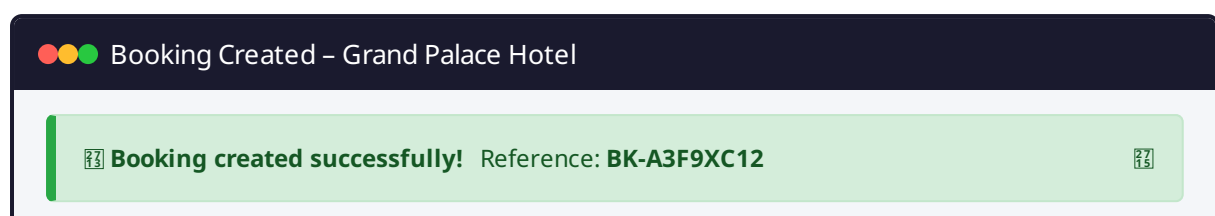


Figure 6.6 – Success notification with auto-generated booking reference after reservation creation

7. CONCLUSION

7.1 Summary

This project successfully demonstrates the design and implementation of a fully functional, production-ready **Hotel and Resort Booking Management System** using **PHP 8.3** and the **Laravel 13.x** framework. The system achieves all primary and secondary objectives outlined at the beginning of this report.

The application provides hotel management staff with a powerful, centralised platform to handle the complete lifecycle of every reservation — from the initial booking request all the way through confirmation, check-in, check-out and, where necessary, cancellation. The five-state status workflow with automatic side effects (room status updates, payment status changes, timestamp recording) ensures that the system's data is always internally consistent, removing the risk of double-bookings or outdated availability information.

The technical implementation showcases the practical power of the Laravel ecosystem: Eloquent ORM relationships and query scopes reduce repetitive database code to clean, expressive PHP; the database transaction wrapper around the booking creation process guarantees atomicity across two related tables; the boot() model hook auto-generates unique reference codes without any controller intervention; and the soft-delete trait preserves historical booking data for auditing purposes even after a record is "deleted" by staff.

The Bootstrap 5 front-end delivers a professional, responsive user interface with an intuitive dark sidebar navigation, colour-coded status badges, gradient statistics cards, and a clean form layout — making the system accessible to non-technical hotel staff without any training overhead.

Key Achievements

- 📦 Complete MVC architecture with clean separation of concerns
- 📦 Real-time room availability with date-overlap conflict detection

-  Atomic booking creation using database transactions
-  Auto-generated unique booking references (BK-XXXXXXX format)
-  Five-stage booking lifecycle with automatic side-effect management
-  Soft-delete data preservation for both rooms and bookings
-  Responsive Bootstrap 5 UI with professional styling
-  Dashboard analytics with live statistics and revenue tracking

7.2 Future Enhancements

While the current system is fully operational and meets all stated requirements, several significant enhancements have been identified for future development phases:

Table 7.1 – Planned Future Enhancements

Enhancement	Description	Priority
Online Payment Gateway	Integration with Stripe or PayPal for secure card payments directly through the booking form	High
Email Notifications	Automated confirmation, reminder and cancellation emails to guests using Laravel Mail with queued jobs	High
Role-Based Access Control	Separate roles for Admin, Receptionist and Manager with granular permissions using Laravel Policies/Gates	High
Guest Self-Service Portal	A guest-facing booking portal where customers can search availability, make reservations and track their own bookings	Medium
Multi-Property Support	Extend the data model to support multiple hotels or resorts under one management platform	Medium

Reporting and Export	PDF invoice generation per booking and Excel/CSV export of booking lists and revenue reports	Medium
Real-Time Notifications	WebSocket-based live notifications (via Laravel Echo and Pusher) when new bookings arrive or statuses change	Low
Housekeeping Module	Maintenance request tracking and room cleaning schedules integrated with the room status system	Low
Mobile Application	React Native or Flutter mobile app consuming a Laravel API back-end for on-the-go management	Low
Advanced Analytics	Interactive revenue charts, occupancy rate graphs, booking source tracking and seasonal demand analysis	Low

7.3 References

1. Laravel Documentation – Official Framework Reference. *laravel.com/docs*. The Laravel Team, 2024. [Accessed May 2026]
2. Taylor Otwell. *Laravel: Up & Running*. O'Reilly Media, 3rd Edition, 2023.
3. Bootstrap 5 Documentation. *getbootstrap.com/docs/5.3*. The Bootstrap Team, 2023.
4. PHP 8.3 Manual. *php.net/manual*. The PHP Group, 2024.
5. Eloquent ORM Documentation. *laravel.com/docs/eloquent*. The Laravel Team, 2024.
6. Carbon – A simple PHP API extension for DateTime. *carbon.nesbot.com*. Brian Nesbitt, 2024.
7. MySQL 8.0 Reference Manual. *dev.mysql.com/doc/refman/8.0*. Oracle Corporation, 2024.
8. Font Awesome 6 Icons. *fontawesome.com*. Fonticons Inc., 2024.

9. Vite – Next Generation Frontend Tooling. *vitejs.dev*. Evan You, 2024.
10. PHPUnit Documentation. *phpunit.de/documentation.html*. Sebastian Bergmann, 2024.

— End of Report —

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